Economic Value of Visual Design: Evidence from the UK Automobile Market

Ankit Sisodia
Purdue University, asisodia@purdue.edu

Vineet Kumar
Yale School of Management, vineet.kumar@yale.edu

March 2026

We examine whether and how visual design of products has economically meaningful effects on demand and competition, focusing on the automobile market. Visual design has been challenging to empirically characterize in a manner that is human-interpretable yet sufficiently rich. We leverage a recently developed method to help achieve low-dimensional disentangled representations of visual design, using data from the UK automobile market across 2008-2017. We obtain human-interpretable visual characteristics that are orthogonal and can reconstruct the original design of all the products (make-models), thus spanning the range of products in the market. We incorporate these visual characteristics into a structural model of demand and supply based on the classic BLP framework. We quantify how economically meaningful quantities like elasticity vary across models with visual characteristics compared with the standard baseline without visual characteristics, and find a substantial economic impact. We also evaluate a counterfactual in which we remove a focal product's closest neighbor in functional (or visual) characteristics, and then comparing how the equilibrium quantities (prices, market shares) vary. We find that removing the closest "visual neighbor" has an effect similar to removing the closest "functional neighbor," highlighting the economically meaningful role of visual characteristics.

Key words: visual analytics, deep learning, demand models

1. Introduction

Visual design or product aesthetics is an important, and even critical factor driving consumer preferences, consideration and choices across a wide range of product categories, ranging from fashion to automobiles to houses. A large body of work in marketing and design highlights that product appearance shapes consumer responses and can be pivotal for market success (Veryzer Jr 1993, Bloch 1995, Creusen and Schoormans 2005, Norman 2013). Firms understand this value, and have typically hired an array of designers to test and prototype different visual designs of products in an attempt to identify and align their designs with consumer preferences (Bloch 1995, Homburg et al. 2015). Firms also choose various functional characteristics for their products in competitive market settings with differentiated products. The issue of new products with functional characteristics and their value to consumers has been studied extensively in the marketing and economics literature (Lancaster 1966, Rosen 1974). Yet, despite the importance, there is very little systematic study of the economic impact of visual design, especially on quantifying the economic value of a new product with a different visual design in real-world market conditions. Consequently, little is known about how to measure the importance of visual design characteristics relative to functional characteristics. Our focus is on quantifying the economic value of visual design and benchmarking it against functional differentiation in a competitive market.

Evaluating the value of new products introduced with a set of characteristics requires the researcher to first specify the characteristics associated with all products. This characteristics-based approach, developed by (Lancaster 1966), makes it possible to model all the products in a market using a small set of typically functional characteristics that essentially serve as a sufficient proxy for consumer preferences. For instance, in the market for fruit juice, the products may be represented by the amount of sugar, tangy taste, calories, organic source etc. For automobiles, the relevant characteristics modeled include engine capability (horsepower), physical dimensions and capacity, fuel efficiency etc. Researchers can then examine a variety of questions like substitution, e.g. when orange juice is out of stock, whether consumers switch to apple juice or grapefruit juice is based on both the characteristics of the products, and consumer preferences over the space of characteristics. Similarly, when a new product is introduced, understanding what the new

equilibrium prices requires a consumer model based on characteristics and preferences across this space.

A noteworthy issue here is that these functional product characteristics, whether calories or horsepower, are all human-interpretable. However, with visual characteristics, such an approach has not been tenable because the basic problem of quantifying visual characteristics in a human-interpretable yet sufficiently rich and useful space of characteristics has been challenging. Related work uses deep learning to map images to interpretable perceptual brand attributes and measure brand portrayal on social media (Liu et al. 2020). Our focus differs in that we learn product-form visual characteristics for use in an equilibrium structural model of demand and supply. The recent approach by Sisodia et al. (2025), building on the disentanglement literature in machine learning, and specifically using functional or structured characteristics like brand and price to learn visual characteristics from product images, provides a promising and systematic approach developed from first principles to help with this problem. The advantage of this disentanglement-based method is that the discovered visual characteristics are typically orthogonal, and the method also aims to provide a low-dimensional representation that can easily be understood even by those not expert in the product domain. The method seeks to provide a one-to-one mapping between the discovered latent factors and the data generating process underlying the visual designs in the data. We also find that the human-interpretable aspect is critical in using judgment to determine which subset of visual characteristics to include in the demand model.

We aim to evaluate the economic value of visual designs in the automobile market in the UK as our empirical setting. We use a combination of structured and unstructured data, with the structured data including typical functional characteristics like brand, horsepower, price and fuel efficiency. The unstructured data includes product images, specifically the front view or "face" of the automobiles in the market. We standardize these images, and using the approach of Sisodia et al. (2025), obtain a small set of human-interpretable visual characteristics. We use the functional characteristics as supervisory signals to overcome the "impossibility theorem," which implies that using only product images cannot help us obtain disentangled visual characteristics.

Our analysis proceeds in several steps. In the first step, we obtain disentangled visual characteristics for use in our downstream problem, which involves a structural equilibrium model. In the second step, we use human judgment to evaluate which of these visual

characteristics are relevant for the downstream application. This is a critical step, since the disentanglement model might discover characteristics that are relevant from the visual product images, but should not be included in the structural model. Consider the role of product color. Typically, the obtained product image data contains a randomly drawn subset of colors for each of the product models. However, there are two reasons why this is not appropriate to use in the structural model. First, automobiles are available in a wide variety of colors, most of which are typically not even available in the product image dataset. Second, market outcomes in data that are used with the structural model are captured at the level of a product make-model, not at the level of individual color. Therefore, we would obtain spurious results if we incorporate such variables, and researcher judgment is helpful, perhaps critical, in choosing the appropriate set of product characteristics. In the third step, we incorporate these visual characteristics within a standard model of demand and supply, based on the standard framework introduced by Berry et al. (1995). We incorporate the discovered visual characteristics directly in the consumer model, where they enter the indirect utility function. We therefore treat them similar to functional characteristics, which are also included. We allow for consumer heterogeneity in valuation of both visual and functional characteristics, allowing for richer patterns of substitution.

We provide an overview of the results, starting with the disentanglement model first. The model discovers four orthogonal visual characteristics. We name these four discovered visual characteristics as *body shape*, *boxiness*, *grille width* and *grille height*. We are then able to map each car make-model to a point in the space of these visual characteristics. The visual characteristics are orthogonal, implying that it is possible to traverse in one direction in the space while only varying one human-interpretable characteristic.

Moving to the structural model, for functional characteristics we find that consumers have overall positive average evaluations for horsepower-to-weight, fuel-cost–adjusted efficiency (MP£), and space. For visual characteristics, we find that consumers have positive preferences on average for body shape and grille height, while boxiness enters negatively. This means that, on average, consumers place value on specific front-end design features above and beyond performance, operating costs, and size, and that substitution patterns depend not only on similarity in functional attributes but also on similarity in visual design preferences. We also find substantial heterogeneity in taste for boxiness and grille height. We then determine the importance of visual characteristics by comparing economic

quantities of interest for a model with visual and functional characteristics compared to the baseline standard model with only functional characteristics. We find that the distribution of own-price elasticities shifts left when visual characteristics are included in the model: the mean own-price elasticity changes from -4.61 (without visual characteristics) to -6.10 (with visual characteristics). This result highlights that visual characteristics are economically meaningful and can impact consumer preferences for products, and consequent elasticities as well.

Our research has been one of the first in incorporating visual product characteristics into a model of demand and supply in the automobile market. We compare how the results are different when we do not include the discovered visual characteristics. We find that incorporating visual characteristics (i) leads to more price-elastic demand (more negative own-price elasticities) and (ii) implies lower markups (mean markup shifts from 0.24 without visuals to 0.18 with visuals), and (iii) changes the model-implied reallocation of demand in equilibrium counterfactuals (e.g., larger predicted share/profit gains following a competitor exit).

One of the important questions is to evaluate the value of visual design, and how it compares to changes in functional characteristics. We quantify this by conducting a counterfactual analysis that compares the competitive effects of removing a product's nearest neighbor in functional space versus its nearest neighbor in visual space. Concretely, for each product in a given market-year, we remove (i) its closest functional neighbor (based on distance in horsepower-to-weight, MP£, and space) and (ii) its closest visual neighbor (based on distance in the learned visual-characteristic space), and then re-solve for the Nash–Bertrand pricing equilibrium holding marginal costs fixed. On average, removing the nearest visual neighbor generates changes in equilibrium outcomes that are similar in order of magnitude to removing the nearest functional neighbor: prices move very little on average (0.009% vs. 0.004%), while market share and profit rise by comparable amounts (0.639% and 0.696% under visual-neighbor removal, versus 0.556% and 0.576% under functional-neighbor removal). These results suggest that differentiating away from competitors in visual space can soften direct competition in a manner comparable to differentiation in performance, operating costs, or size.

Our primary contribution is in empirically evaluating and quantifying the impact of visual design. We do this by automatically discovering human-interpretable visual characteristics of products in the automobile market, and incorporating them in an equilibrium

structural model to quantify how visual design affects consumer preferences, substitution patterns, and equilibrium market outcomes. We expect this research to open up an avenue for further exploration, including studying the evolution of designs over time, the competitive dynamics of visual design and visual differentiation in different market settings.

2. Literature Review

This paper quantifies how visual product design affects demand and competitive interactions in a differentiated-products market. We build on four streams of literature: (i) marketing research on product aesthetics and design, (ii) disentangled representation learning for extracting interpretable factors from images, (iii) structural demand and supply estimation in the BLP tradition, and (iv) methods that incorporate unstructured data into discrete-choice models. We expand on these below.

2.1. Visual Product Design

A longstanding literature in marketing argues that product form influences consumer preference and choice beyond observed functional attributes. Bloch (1995) proposes that product form generates cognitive and affective responses that affect approach and avoidance behavior. Related research highlights multiple roles of appearance in consumer choice, including aesthetic value, symbolic meaning, and communication of functional information (Creusen and Schoormans 2005, Buechel and Townsend 2018). In the automotive context, Kang et al. (2019) shows that consumers make explicit trade-offs between visual form and functional attributes, and related work on design typicality and novelty suggests that relative visual positioning shapes consumer evaluation (Talke et al. 2009). A remaining challenge is to translate these insights into market-level estimates of how specific, measurable design dimensions shape substitution and competitive outcomes in equilibrium. Doing so requires mapping visual design into a set of product characteristics that can enter a demand system and support counterfactuals.

Despite this evidence, incorporating visual design directly into structural demand models has been challenging because visual design is inherently unstructured and difficult to quantify using standard product databases. Moreover, for structural estimation and interpretation, any measured visual characteristic must be interpretable in substantive terms, i.e., it should be possible to understand what a unit change represents. As a result, visual design is often absorbed into the unobserved component of utility, which can matter for demand estimation and inference when correlated with endogenous variables such as price.

This motivates approaches that transform product images into structured and interpretable product characteristics that can enter the utility specification and be valued within a structural model of demand.

2.2. Disentangled Representation Learning

Representation learning studies how high-dimensional observations can be encoded using lower-dimensional factors that retain information useful for downstream tasks (Bengio et al. 2013). This paper focuses on a branch called disentangled representation learning, which aims to isolate meaningful, independent factors of variation in data (Bengio et al. 2013). Disentanglement seeks to obtain low-dimensional and human-interpretable factors of variation from high-dimensional data.

A large class of disentanglement approaches is based on variational autoencoders (VAEs), including β-VAE (Higgins et al. 2017, Burgess et al. 2017), FactorVAE (Kim and Mnih 2018), and β-TCVAE (Chen et al. 2018). However, a key challenge is that in purely unsupervised settings there is no general theoretical guarantee that a learned representation is uniquely disentangled (Locatello et al. 2019). This result is sometimes described as an "impossibility theorem" for unsupervised disentanglement. Learning semantically interpretable and stable latent dimensions typically requires some form of inductive bias or supervision. Locatello et al. (2020) showed that even limited, potentially imperfect supervision can be sufficient for model selection when learning disentangled representations.

Building on this insight, Sisodia et al. (2025) demonstrated that product characteristics available in standard marketing datasets (e.g., brand, price, physical dimensions) can provide useful supervisory signals that help learn and select disentangled representations for product images without pre-specifying the visual factors to recover, or requiring manual identification and labeling of visual attributes. The learned disentangled representations can then be validated post hoc to ensure that they are human-interpretable. In our setting, this emphasis on interpretability is important because the learned dimensions will be treated as product characteristics in a structural model and used for counterfactual analysis.

2.3. Demand Models

The random coefficient logit model described in Berry et al. (1995) is a foundational framework for estimating demand for differentiated products in economics and marketing. The

model posits that consumer utility depends on observed product characteristics available to the researcher as well as unobserved product characteristics observed by market participants but not by the researcher. These unobservables can be correlated with price, motivating the instrumental-variable approach, following Berry (1994). By allowing heterogeneous tastes over product characteristics, the framework yields flexible substitution patterns and is well-suited to settings with aggregate market-level data and many differentiated products.

Subsequent work has extended the BLP framework along several dimensions. Nevo (2000) incorporated interactions between observed consumer demographics and product characteristics and showed how to include product fixed effects. Petrin (2002) and Berry et al. (2004) integrated micro data with market-level data, yielding richer substitution patterns and reducing instrument requirements. These extensions have improved how researchers model consumer heterogeneity and control for unobservables. However, the set of product characteristics entering the demand model has remained limited to variables that can be recorded as structured data. Relevant design variation is visible in images but not directly available as structured characteristics, motivating methods that convert images into usable product attributes.

2.4. Unstructured Data in Consumer Choice Models

A growing literature in marketing and economics leverages unstructured data, such as product images, titles, descriptions, and customer reviews, to enrich discrete choice models. A common approach is to transform unstructured inputs into low-dimensional representations (embeddings) using pre-trained deep learning models, reduce dimensionality using methods such as principal components analysis, and incorporate the resulting components as additional product characteristics in demand models. This work emphasizes that unstructured data can capture otherwise hard-to-quantify dimensions of differentiation and, in turn, can improve the model's ability to recover substitution patterns relevant for counterfactual analyses.

Within this literature, several papers use unstructured data primarily to predict mean utilities or product intercepts. For example, Quan and Williams (2019) incorporate product image embeddings from pre-trained convolutional neural networks as shifters of mean utility intercepts, showing that visual information helps explain cross-sectional variation in product demand. Similarly, Lee (2025) uses large language models to predict utility

intercepts for new products based on their textual descriptions. These approaches are valuable for measuring aspects of product positioning that are otherwise unobserved in structured datasets, but they do not directly parameterize the substitution patterns that drive competitive interactions in random-coefficients demand models. Recent work goes beyond mean-utility shifters by incorporating text- and image-based representations directly into random coefficient demand models to capture richer substitution patterns from unstructured data (e.g., Compiani et al. (2025)). While their approach is scalable across categories and does not require pre-specified attributes, the principal components that enter the model are less directly interpretable.

A complementary set of approaches uses survey-based approaches to measure perceived similarity and incorporate it into demand estimation. For instance, Dotson et al. (2016) elicit ratings of product images and use rating correlations to shift utility correlations, while Magnolfi et al. (2025) elicit relative similarity judgments (e.g., triplets) to construct embeddings that enter a random coefficient logit model. These methods can provide direct information about perceived similarity but typically require category-specific data collection.

Our Contribution:

This paper contributes to the literature by proposing an approach that is both interpretable and structural. In fact, developing a structural model requires that all the elements that impact consumer preferences (and choices) be human interpretable and based on first principles. This requirement directly motivates our approach. Rather than using opaque embeddings or principal components as product characteristics, we extract interpretable visual dimensions from product images using disentangled representation learning and incorporate them as additional observed product attributes in a BLP demand model. Our approach complements embedding-based structural methods by recovering a small set of interpretable visual dimensions that enter utility directly as product attributes. In doing so, it shifts systematic measured visual variation from the unobserved component ξ_{jt} into the utility specification, making it available for preference estimation and counterfactual analysis.

3. Methodology

We propose a two-stage approach to incorporate interpretable visual product characteristics into a structural model of demand and competition. In the first stage, we learn a

low-dimensional, disentangled representation of product images using a variational autoencoder (VAE) with weak supervision from product characteristics (Sisodia et al. 2025) found in structured data. In the second stage, we estimate a random-coefficients logit demand model in the BLP class (Berry et al. 1995) and a supply-side pricing model, treating the learned visual characteristics as additional observed product characteristics. Endogeneity of price is addressed using instrumental variables.

3.1. Learning Visual Characteristics from Images

Our approach to discovering visual product characteristics follows Sisodia et al. (2025) and employs Variational Autoencoders (VAEs) (Kingma and Welling 2014) for disentangled representation learning (Bengio et al. 2013, Higgins et al. 2017, Burgess et al. 2017, Chen et al. 2018, Kim and Mnih 2018). A VAE consists of (i) an encoder $q_\phi(\mathbf{h} \mid \mathcal{I})$ that maps each high-dimensional product image $\mathcal{I}$ to a low-dimensional latent vector $\mathbf{h} \in \mathbb{R}^D$ and (ii) a decoder $p_\theta(\mathcal{I} \mid \mathbf{h})$ that reconstructs the image from $\mathbf{h}$. We assume a standard Gaussian prior $p(\mathbf{h}) = \mathcal{N}(0, I)$, and a Gaussian variational posterior $q_\phi(\mathbf{h} \mid \mathcal{I}) = \mathcal{N}(\mu_\phi(\mathcal{I}), \mathrm{diag}(\sigma^2_\phi(\mathcal{I})))$. By training the encoder and decoder jointly, the model learns a compact set of latent factors $\mathbf{h}$ that capture the salient dimensions of variation in product images. Our goal is to obtain a low-dimensional set of visual characteristics that summarizes the main dimensions of variation across product images and can be used as product characteristics in the demand model. Architectural and training details are provided in Appendix A.

Why is supervision needed? A core challenge is that in purely unsupervised settings there is no general theoretical guarantee that the learned representation is uniquely disentangled (Locatello et al. 2019). In fact, a true disentangled representation can be probabilistically equivalent to an infinite set of entangled representations, a result termed as the "impossibility theorem." In practice, learning stable, human-interpretable factors typically requires inductive biases and/or some form of supervision (Locatello et al. 2020). Much of the literature in machine learning assumes access to labels for at least part of the data (often termed weak supervision). In our setting, the ground-truth visual attributes are precisely what we would want the model to automatically discover, and by definition are unavailable. To get around this impossibility theorem, we use product characteristics found in structured data as supervisory signals, following Sisodia et al. (2025). Intuitively, some functional or structured characteristics (e.g., brand, price, and other observed product attributes) are likely to be correlated with underlying design elements and can guide the latent space

toward meaningful visual variation. We emphasize that these supervisory signals are not treated as ground-truth labels; rather, they serve as correlated proxies that help orient the learned latent factors.

Objective function. We use the β-TCVAE framework (Chen et al. 2018) augmented with a supervised loss. Let $\mathbf{y}$ denote a vector of supervisory signals associated with image $\mathcal{I}$ and let $\widehat{\mathbf{y}}(\mathbf{h})$ denote predictions from a supervised head that takes the latent vector $\mathbf{h}$ as input. The learning objective is:

$$\underbrace{\mathcal{L}(\theta,\phi;\mathcal{I},\mathbf{h})}_{\text{Disentanglement Loss}} = \underbrace{-\mathbf{E}_{q_\phi(\mathbf{h}|\mathcal{I})}[\log p_\theta(\mathcal{I}\mid\mathbf{h})]}_{\substack{\text{Reconstruction}\\ \text{Loss}}} + \underbrace{I_q(\mathbf{h},\mathcal{I})}_{\substack{\text{Mutual}\\ \text{Information}\\ \text{Loss}}} + \beta \underbrace{\mathrm{KL}\left[q(\mathbf{h})\,\middle\|\,\prod_{d=1}^{D} q(h_d)\right]}_{\substack{\text{Total Correlation}\\ \text{Loss}}}$$
$$+ \underbrace{\sum_{d=1}^{D}\mathrm{KL}[q(h_d)\,\|\,p(h_d)]}_{\substack{\text{Dimension-Wise}\\ \text{KL Divergence Loss}}} + \delta \underbrace{P(\widehat{\mathbf{y}}(\mathbf{h}),\mathbf{y})}_{\substack{\text{Supervised}\\ \text{Loss}}} \tag{1}$$

Here $q(\mathbf{h})$ and $q(h_d)$ denote the aggregate posterior and its marginals. The hyperparameter β controls the strength of the disentanglement penalty (via total correlation), while δ controls the strength of weak supervision. Intuitively, the reconstruction and mutual-information terms ensure $\mathbf{h}$ remains informative about the input image, the total-correlation term discourages dependence across latent dimensions, the dimension-wise KL term regularizes each latent dimension toward its prior, and the supervised term encourages alignment between the learned representation and product characteristics found in structured data that are correlated with design.[1] Large values of δ can overweight predictive fit relative to disentanglement, while $\delta = 0$ removes the proxy supervision needed to address the impossibility result (Locatello et al. 2019).

Hyperparameter and model selection. We consider a pre-specified menu $\mathcal{Y}$ of candidate supervisory-signal specifications constructed from product characteristics found in structured data (defined in Section 4.2). For each $\mathbf{y} \in \mathcal{Y}$, we tune (β, δ) by grid search and select the pair that minimizes 10-fold cross-validated supervised loss (Locatello et al. 2020). We then refit the model under the selected (β, δ) for each candidate $\mathbf{y}$ and select $\mathbf{y}^\star$ as the supervisory-signal specification that yields the highest Unsupervised Disentanglement

[1] For categorical supervisory signals, the supervised loss is cross-entropy, and for continuous supervisory signals, the supervised loss is mean squared error.

Ranking (UDR) score (Duan et al. 2020). UDR evaluates robustness of the learned representation across random initializations without requiring ground-truth labels. It relies on the assumption that a well-disentangling VAE will converge on the same representation up to isomorphic transformations. The UDR algorithm is detailed in Appendix B.

Table 1 Table of Notation

Symbol	Meaning
Indices and Sets	
j	Products ($j = 0, 1, \ldots, J_t$)
t	Markets
i	Consumers
f	Firms
T	Number of markets
J_t	Number of inside products in market t (excluding $j = 0$)
k_1, k_2	Indices for elements of $\mathbf{x}_{jt}^D$ and $\mathbf{v}_{jt}$
I_t	Number of consumers in market t
F_t	Number of firms in market t
$\mathcal{F}_f(t)$	Set of products owned by firm f in market t
Observables	
p_{jt}	Price (inside goods $j = 1, \ldots, J_t$)
$\mathbf{x}_{jt}^D$	Functional (non-visual) characteristics in demand mean utility
$\mathbf{x}_{jt}^C$	Cost-side characteristics entering marginal cost
$\mathbf{v}_{jt}$	Visual product-characteristics vector (learned via VAE)
x_{jt}^{D,k_1}	k_1th element of $\mathbf{x}_{jt}^D$
$v_{jt}^{k_2}$	k_2th element of $\mathbf{v}_{jt}$
y_{it}	Income of consumer i in market t
$\mathbf{w}_{jt}$	Observed cost-shifter vector (enters marginal cost)
z_{jt}^{cost}	External product-specific cost shifter (lagged)
Utility Components	
U_{ijt}	Indirect utility of consumer i for product j in market t
δ_{jt}	Mean utility
μ_{ijt}	Individual-specific deviation from mean utility
ϵ_{ijt}	Idiosyncratic taste shock (Type I extreme value)
ν_{it}	Taste draws (vector; standard normal)
ν_{it}^k	kth element of ν_{it}
δ_{0t}	Outside-option mean utility (normalized to 0)
μ_{i0t}	Outside-option deviation (normalized to 0)
Parameters	
ζ	Full parameter vector
$\bar{\boldsymbol{\beta}}_1$	Vector of mean preferences for $\mathbf{x}_{jt}^D$
$\bar{\boldsymbol{\beta}}_2$	Vector of mean preferences for $\mathbf{v}_{jt}$
σ_{β_1}	Vector of std. devs. for random coefficients on functional characteristics
σ_{β_2}	Vector of std. devs. for random coefficients on visual characteristics
α	Price sensitivity parameter
γ_1, γ_2	Vectors of marginal cost parameters
Choice and Shares	
s_{ijt}	Individual choice probability
s_{jt}	Market share
Unobservables	
ξ_{jt}	Demand-side unobserved product–market characteristic
ω_{jt}	Supply-side unobserved cost shock
c_{jt}	Marginal cost
Estimation	
Z_{jt}^D	Demand instrument vector
Z_{jt}^S	Supply instrument vector
N	Number of inside product–market observations ($N = \sum_{t=1}^T J_t$)
W	GMM weighting matrix
$g(\zeta)$	Sample moment function

Constructing $\mathbf{v}_{jt}$. After training under the selected specification $\mathbf{y}^\star$, the encoder yields a latent vector $\mathbf{h}$ for each product image $\mathcal{I}$. We construct $\mathbf{v}_{jt}$ in three steps. First, we retain as *informative* those latent dimensions whose KL divergence from the prior exceeds a threshold (standard practice in the disentanglement literature). Second, we interpret the informative dimensions using latent traversals. Third, we retain the subset corresponding to design-related visual factors and enter these dimensions into the demand model as additional observed product characteristics.

3.2. Demand and Supply Model

We estimate a model of market equilibrium using a random-coefficients logit demand system in the BLP class (Berry et al. 1995). Table 1 presents the notation used in the model.

Consumers. In each market $t = 1, \ldots, T$, consumers choose among J_t differentiated products $j = 1, \ldots, J_t$ and an outside option $j = 0$. Each consumer has unit demand and chooses the alternative with highest utility among $\{0, 1, \ldots, J_t\}$, where the utility of the outside option (0) is normalized by setting $\delta_{0t} = 0$ and $\mu_{i0t} = 0$. We observe prices p_{jt}, functional (non-visual) characteristics $\mathbf{x}^D_{jt}$, learned visual characteristics $\mathbf{v}_{jt}$ (Section 3.1), and market shares for all products.

Consumer i's indirect utility from product j in market t is:

$$U_{ijt} = \delta_{jt} + \mu_{ijt} + \epsilon_{ijt}, \tag{2a}$$

$$\delta_{jt} = (\mathbf{x}^D_{jt})'\bar{\boldsymbol{\beta}}_1 + \mathbf{v}'_{jt}\bar{\boldsymbol{\beta}}_2 + \xi_{jt}, \tag{2b}$$

$$\mu_{ijt} = -\alpha \frac{p_{jt}}{y_{it}} + \sum_{k_1} \sigma^{k_1}_{\beta_1} x^{D,k_1}_{jt} \nu^{k_1}_{it} + \sum_{k_2} \sigma^{k_2}_{\beta_2} v^{k_2}_{jt} \nu^{k_2}_{it}. \tag{2c}$$

Here y_{it} is consumer income, $\nu_{it} = (\nu^k_{it})_k$ denote consumer i's vector of taste draws in market t (assumed mean-zero standard normal), and ϵ_{ijt} is i.i.d. Type-I extreme value. The term ξ_{jt} captures demand-relevant product–market unobservables not explained by $(\mathbf{x}^D_{jt}, \mathbf{v}_{jt})$. We set $\sigma_{\text{price}} = 0$ so that heterogeneity in price sensitivity operates through observable income variation via $-\alpha p_{jt}/y_{it}$.

Shares. Under the Type-I extreme value assumption, the individual choice probabilities are:

$$s_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{\sum_{\ell \in \{0,1,\ldots,J_t\}} \exp(\delta_{\ell t} + \mu_{i\ell t})}, \qquad s_{jt} = \int s_{ijt}\, dF(\nu, y), \tag{3}$$

where for the outside option we normalize $\delta_{0t} = 0$ and $\mu_{i0t} = 0$. We approximate the integral by simulation (1000 scrambled Halton draws per market).

Firms (Supply). In each market t, firms play a static, full-information Nash–Bertrand pricing game. Each multi-product firm $f \in F_t$ chooses the prices of all its products in market t to maximize total profit. Let $\mathcal{F}_f(t)$ denote the set of products owned by firm f in market t.

Firms set prices by solving the first order condition in prices. The first-order condition for product $j \in \mathcal{F}_f(t)$ is:

$$s_{jt} + \sum_{r \in \mathcal{F}_f(t)} (p_{rt} - c_{rt}) \frac{\partial s_{rt}}{\partial p_{jt}} = 0, \tag{4}$$

where the summation captures multi-product internalization: the firm accounts for cross-price effects among its own products when setting p_{jt}.

We parameterize log marginal costs as:

$$\ln(c_{jt}) = (\mathbf{x}_{jt}^C)'\gamma_1 + \mathbf{w}_{jt}'\gamma_2 + \omega_{jt}, \tag{5}$$

where $\mathbf{x}_{jt}^C$ are cost-side product characteristics, $\mathbf{w}_{jt}$ are observed cost shifters, and ω_{jt} is an unobserved cost shock. Given estimated demand derivatives $\partial s_{rt}/\partial p_{jt}$, equation (4) implies markups, which we use to back out marginal costs.

We do not include visual characteristics $\mathbf{v}_{jt}$ directly in Equation (5). Conditional on $\mathbf{x}_{jt}^C$, any cost-side variation correlated with visual design is absorbed by ω_{jt}. We therefore interpret $\mathbf{v}_{jt}$ primarily as demand shifters rather than cost shifters in the baseline specification.

3.3. Instruments and Estimation

Moment conditions. Price is endogenous because firms observe ξ_{jt} when setting prices p_{jt}, generating correlation between price p_{jt} and the structural error ξ_{jt}. We maintain the standard full-information pricing assumption that firms observe the full vector of demand unobservables when setting prices, while the econometrician does not. We address this using instrumental variables.

Let Z_{jt}^D denote demand instruments and Z_{jt}^S denote supply instruments. We use the following moment conditions:

$$\mathbb{E}[\xi_{jt} Z_{jt}^D] = 0, \qquad \mathbb{E}[\omega_{jt} Z_{jt}^S] = 0. \tag{6}$$

Instruments. We use three classes of instruments.

(i) BLP instruments. First, following Berry et al. (1995), we construct instruments from (a) counts of other products offered by the same firm and by rival firms in market t, and (b) sums of other products' observed demand characteristics. We form these instruments separately for own-firm and rival products, excluding product j from own-firm sums. For the demand system, the characteristic sums use elements of $\mathbf{x}_{jt}^D$ and $\mathbf{v}_{jt}$. For the supply system, we construct own-firm and rival sums of elements of $\mathbf{x}_{jt}^C$ and $\mathbf{w}_{jt}$.

These instruments shift equilibrium prices through variation in the competitive environment (demand-side sums/counts) and through variation in cost shifters (supply-side sums). Identification requires that the demand instruments are orthogonal to the demand unobservable ξ_{jt} and that the supply instruments are orthogonal to the cost shock ω_{jt}, conditional on the included controls and under the standard timing assumption that characteristics and cost shifters are predetermined by the time prices are set.

(ii) Local (segment / region) variants. Second, following the logic of Sudhir (2001), we construct "local" variants of the instruments in (i) by restricting the counts and sums to products within the same market segment and, separately, within the same production region. These local instruments add variation from competition among closer substitutes and from groups with more similar cost structures (Gandhi and Houde 2019).

(iii) External cost shifter. In addition to the characteristic-based instruments, we use an external, product-specific cost shifter that varies by the product's production location and time. Let z_{jt}^{cost} denote this shifter, measured in the production location associated with product j and lagged one period. The identifying assumption is that z_{jt}^{cost} shifts equilibrium prices through marginal costs (e.g., via exchange-rate movements) but is orthogonal to the demand unobservable ξ_{jt} (and to the cost shock ω_{jt} when used for supply-side moments) in market t, conditional on the included covariates in the demand and marginal-cost equations. We therefore include z_{jt}^{cost} in both the demand and supply instrument sets. We defer details on z_{jt}^{cost} to Section 4.1.

GMM estimation. Let ζ collect all parameters in the demand and supply system. Define the sample moments

$$g(\zeta) = \begin{bmatrix} \frac{1}{N}\sum_{j,t} \xi_{jt}(\zeta)\, Z_{jt}^D \\ \frac{1}{N}\sum_{j,t} \omega_{jt}(\zeta)\, Z_{jt}^S \end{bmatrix}, \tag{7}$$

where $\xi_{jt}(\zeta)$ and $\omega_{jt}(\zeta)$ are the demand-side and supply-side residuals implied by parameter values ζ, and $N = \sum_t J_t$ is the number of product–market observations. We estimate ζ by minimizing

$$\widehat{\zeta} = \arg\min_{\zeta} \; g(\zeta)^\top W \, g(\zeta), \tag{8}$$

where W is a weighting matrix. We implement a two-step GMM procedure: we first obtain consistent estimates using an initial W, then update W using the estimated moment covariance matrix.

Following Conlon and Gortmaker (2020), we also implement a two-step procedure with approximate optimal instruments: (i) estimate the model using the constructed instruments to obtain consistent estimates $\widehat{\zeta}^{(1)}$; (ii) use $\widehat{\zeta}^{(1)}$ to form approximate optimal instruments based on the expected Jacobian of the residuals with respect to parameters, and re-estimate the model for improved efficiency.

4. Empirical Setting and Results

We focus on the automobile industry in the United Kingdom (UK) from 2008 to 2017. Visual design is an important factor in consumer purchase decisions in the automobile market. Industry evidence underscores this importance in the consumer decision process, including in consideration (or avoidance), with the J.D. Power Avoider Study (2015) reporting:

> "Exterior look/design is the top reason shoppers avoid a particular vehicle (30%), followed by cost (17%)."

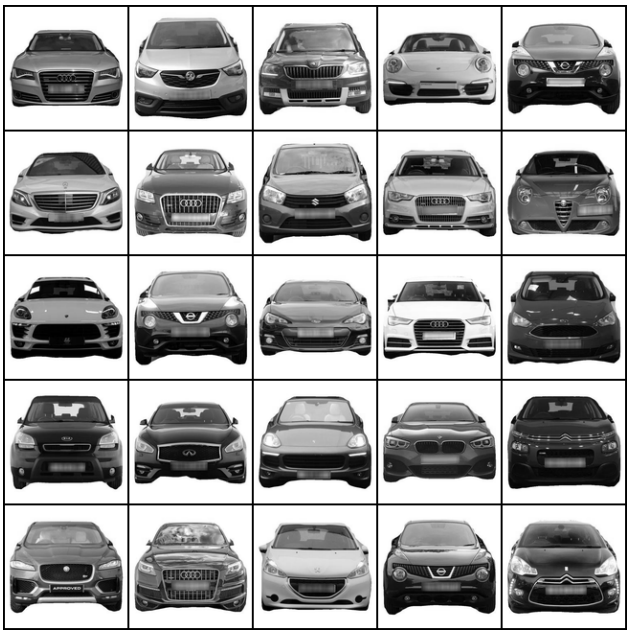
We next describe our dataset, which combines automobile characteristics, market shares, and product images. Next, we discuss the visual characteristics discovered through our disentangled representation learning approach, comparing the performance of supervised and unsupervised methods. To validate both interpretability and quantification of these visual characteristics, we present results from human subject surveys. Finally, we incorporate the discovered visual characteristics into our BLP demand model and report estimation results, including how design factors affect consumer choices.

4.1. Data

We compile a panel of new automobile sales, characteristics, and images in the United Kingdom (UK) covering model-years 2008–2017. The unit of observation in the estimation sample is a make–model–year. We obtain annual new-car quantities and product images

from DVM-CAR (Huang et al. 2021). We select front-facing images only, as industry studies suggest that up to 70% of consumers identify and judge vehicles by the appearance of headlights and grille located on the face of the automobile.[2] We resize each image to 128×128 pixels and convert it to grayscale so that the learned visual characteristics capture shape-related variation rather than color.[3] Figure 1 displays a sample of images from the dataset.

Figure 1 Sample of Automobile Images



We collect manufacturer suggested retail prices (MSRP) and vehicle characteristics from Parker's for all automobiles sold in the UK during 2008–2017. The characteristics include horsepower, CO_2 emissions, weight, and space, measured as length times width. Prices are deflated to 2015 pounds sterling using the UK consumer price index. We supplement these data with each vehicle's primary country of production and firm ownership information, and we obtain additional market-level series from the UK Office for National Statistics (ONS), including the number of households and annual fuel prices for ultra-low sulphur petrol and ultra-low sulphur diesel. Following Berry et al. (1995), we construct miles per pound sterling (MP£) by combining CO_2 emissions, fuel type, and prevailing UK fuel prices to obtain a fuel-cost–adjusted efficiency measure.

[2] See, e.g., https://www.wsj.com/articles/SB114195150869994250.

[3] Another motivation to convert images to grayscale is because we do not have sales at a make-model-year-color level.

Since sales in DVM-CAR are reported at the make–model level, while many vehicle characteristics are available at the trim level, we aggregate trim-level characteristics to the make–model–year by taking the median across trims. We restrict attention to passenger cars with at least 10 registered sales per model-year and exclude battery electric vehicles (EVs), which comprise a negligible share of UK sales in our sample period. The final estimation sample comprises approximately 2,125 model-year observations covering 326 distinct models from 44 makes across 10 years, representing roughly 96% of UK new-car sales over 2008–2017.[4]

We define a market as the UK new automobile market in a given year $t \in \{2008, \ldots, 2017\}$. The potential market size M_t is the total number of UK households in year t, obtained from the ONS. Each household is assumed to purchase at most one new vehicle per year. Let q_{jt} denote the number of new cars sold (registered) for product j in year t. We construct observed inside-good market shares as $s_{jt} = q_{jt}/M_t$, and the outside share as $s_{0t} = 1 - \sum_{j=1}^{J_t} s_{jt}$.

To implement the random-coefficients specification, we require a distribution of consumer income in each market-year. We use annual mean and median real equivalised household disposable income for the UK (ONS). For each year t, we fit a lognormal income distribution by setting $\mu_t = \ln(\text{median}_t)$ and $\sigma_t = \sqrt{2(\ln(\text{mean}_t) - \mu_t)}$. We draw simulated incomes from this fitted distribution when computing individual choice probabilities.

Table 2 reports sales-weighted summary statistics by year for the estimation sample, including quantity of new vehicles sold, price (in £), horsepower, weight (in kg), space (in m^2), CO_2 emissions (in g/km), and MP£(miles/£).

[4] We do not have images for approximately 10% of make–model–years, which collectively account for about 4% of sales during 2008–2017.

Table 2 Summary Statistics by Market

Market	Models	Quantity (units)	Price (£)	HP (HP)	Weight (kg)	Space (m^2)	CO_2 (g/km)	MP£ (miles/£)
2008	189	7411.98	21356.97	119.56	1319.01	8.03	156.80	7.51
2009	204	7151.62	21089.88	114.85	1291.79	7.98	148.05	8.88
2010	209	7529.94	21397.50	117.21	1298.75	8.05	142.44	8.23
2011	203	7227.45	21892.95	124.17	1329.86	8.11	138.48	7.67
2012	217	7436.30	21842.88	124.42	1331.91	8.19	131.64	8.17
2013	212	8657.40	21678.64	123.31	1319.19	8.22	126.66	8.77
2014	221	9107.77	22003.75	124.22	1315.50	8.28	123.65	9.52
2015	222	9518.27	23151.88	127.40	1323.26	8.37	121.30	11.15
2016	231	9712.38	24087.19	130.93	1332.52	8.46	122.26	11.43
2017	217	9070.10	24659.27	134.61	1347.75	8.48	124.16	10.66
All	2125	8323.49	22462.54	124.76	1322.08	8.24	131.83	9.40

Price is endogenous in the demand system and requires instruments. Following Grieco et al. (2021), we use an external cost shifter based on the real exchange rate (RXR) for the country where each vehicle is primarily produced, constructed from the Penn World Table (Feenstra et al. 2015) and lagged one year. The instrument varies across products within a market because vehicles sold in the UK are manufactured in 12 countries. We also construct characteristic-based instruments as described earlier. Details on instrument construction and identifying assumptions appear in Section 3.3.

4.2. Discovered Visual Characteristics

We apply the supervised disentanglement approach described in Section 3.1 to learn visual characteristics from front-facing automobile images. We compare across a variety of supervisory signal specifications, selecting the one that yields the highest Unsupervised Disentanglement Ranking (UDR) score. The UDR score provides a quantitative measure of disentanglement between 0 and 1, with higher values representing more disentangled representations. Details of the UDR algorithm are in Appendix B.

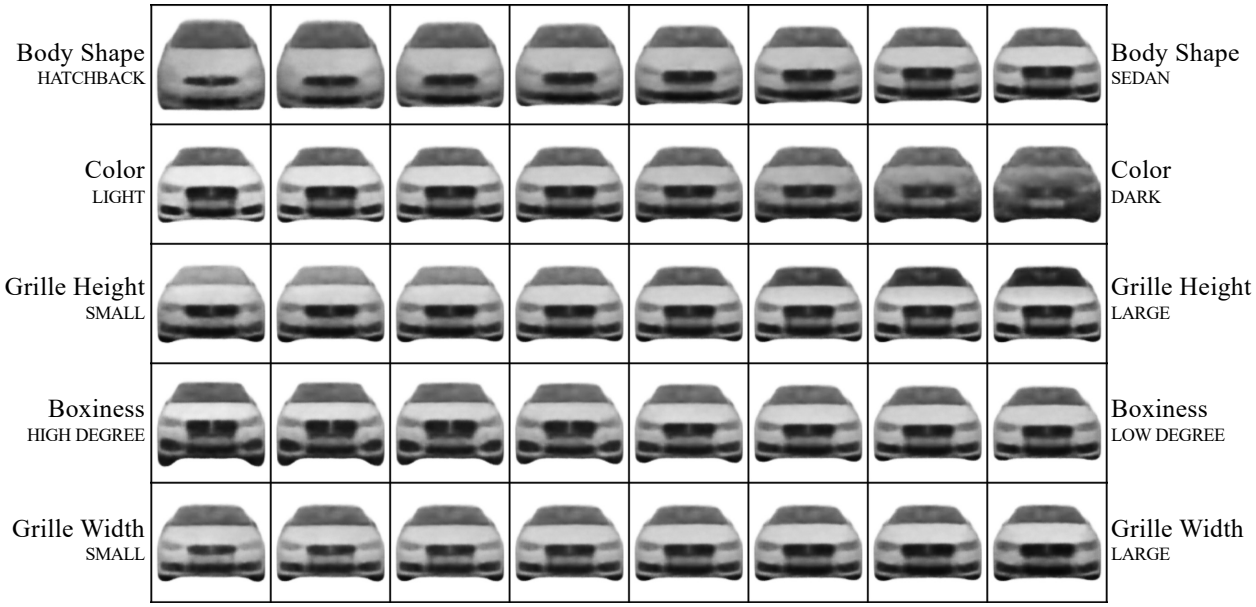
Table 3 reports UDR scores for selected supervisory specifications. The combination of wheelbase, width, and height achieves the best disentanglement (UDR = 0.739), outperforming both single-signal specifications and unsupervised approaches. Unsupervised methods, including β-TCVAE, standard VAE, and autoencoder baselines, yield substantially lower UDR scores, suggesting that weak supervision is helpful to learn stable, interpretable representations. Full results for all candidate specifications appear in Appendix D.

Table 3 Comparison of Different Supervisory Approaches

# Signals	Supervisory Signals	β	δ	UDR
3	Wheelbase, Width, Height	50	10	0.739
1	Price	50	30	0.708
1	Wheelbase	50	30	0.690
1	Width	50	5	0.689
0	Unsupervised β-TCVAE	50	0	0.658
1	Height	30	20	0.378
0	Unsupervised VAE	1	0	0.073
0	Unsupervised AE	0	0	0.074

Notes: UDR lies in $[0,1]$, with larger values indicating more disentangled representations. For each supervisory specification, we report the (β,δ) pair selected via cross-validated supervised loss over the grid $\beta \in \{1,5,10,20,30,40,50\}$ and $\delta \in \{0,1,5,10,20,30,40,50\}$. Unsupervised baselines correspond to $\delta = 0$. Full results appear in Appendix D.

Figure 2 Discovered Visual Characteristics



Notes: Each row varies one visual characteristic from low (left) to high (right) while holding others fixed. Model: β-TCVAE with $\beta = 50$, $\delta = 10$, supervised on (Wheelbase, Width, Height).

Figure 2 displays latent traversals for the selected model ($\beta = 50$, $\delta = 10$, supervised on wheelbase, width, and height). Each row varies one latent dimension while holding others fixed, allowing visual interpretation of what each dimension captures. Throughout, "low" and "high" refer to the leftmost and rightmost images in each traversal row in Figure 2. We

identify four informative visual characteristics and assign labels based on interpretability-survey responses summarized in Appendix C:

1. *Body Shape*: High scores correspond to narrower, more angular profiles (sedan-like); low scores correspond to wider, rounder profiles (hatchback-like).
2. *Boxiness*: Low scores correspond to taller, more upright, narrower silhouettes; high scores correspond to lower, flatter, wider profiles.
3. *Grille Height*: Higher scores indicate a more prominent, vertically extended grille.
4. *Grille Width*: Higher scores indicate a wider, more horizontally extended grille.

A fifth dimension capturing color (light to dark traversal in Figure 2 and not a change in hue) was identified but excluded from the demand model, as this dimension does not correspond to a design attribute. The four visual characteristics above constitute $\mathbf{v}_{jt}$ in the structural model. To assess interpretability and quantification, we conducted two surveys. In the interpretability survey ($N = 93$ after attention checks), respondents viewed five-image traversal sequences for each characteristic (varying one latent dimension while holding others fixed) and described the dominant visual change. In the quantification survey ($N \approx 103$–107 per characteristic), respondents completed pairwise similarity tasks, and agreement with the learned ordering exceeds 90% for all four characteristics. Survey design and detailed results appear in Appendix C.

We examine how the discovered visual characteristics relate to one another, to the functional characteristics used in the structural model, and to basic vehicle dimensions. Table 4 shows that the four visual characteristics are only weakly correlated with one another (maximum $|\rho| = 0.12$), consistent with the disentanglement objective of separating distinct aspects of front-view design.

Table 4 Correlations Among Discovered Visual Characteristics

	Body Shape	Boxiness	Grille Height	Grille Width
Body Shape	—			
Boxiness	0.06	—		
Grille Height	−0.01	0.05	—	
Grille Width	−0.12	−0.01	−0.05	—

Notes: Entries are pairwise correlations in the estimation sample.

Table 5 reports correlations between visual and functional characteristics. Overall, these correlations are generally small to moderate, suggesting that the visual characteristics

are not simple proxies for standard functional characteristics. The strongest relationships involve *Body Shape*, which correlates with HP/Weight and with *Space*, while *Grille Height* and *Grille Width* exhibit near-zero correlations with MP£ and only weak correlations with the other functional characteristics.

Table 5 Correlations: Visual vs. Functional Characteristics

	HP/Weight	MP£	Space
Body Shape	0.49	−0.18	0.47
Boxiness	0.25	0.01	0.16
Grille Height	0.14	0.05	0.02
Grille Width	0.03	−0.02	0.15

Notes: Entries are pairwise correlations in the estimation sample.

Table 6 relates the visual characteristics to physical dimensions. Since lower *Body Shape* scores correspond to hatchback-like fronts while higher scores correspond to sedan-like fronts, the positive correlations with wheelbase, length, and weight imply that sedan-like designs tend to be larger along these dimensions, while hatchback-like designs tend to be smaller. Likewise, its negative correlation with height-to-width ratio implies that cars with lower *Body Shape* scores appear taller and narrower—characteristics commonly associated with hatchbacks or smaller crossover-style cars. *Boxiness* is strongly negatively correlated with vehicle height and with the height-to-width ratio, indicating that cars with higher degree of boxiness (lower score on the visual characteristic of boxiness) tend to look taller and more upright from the front. In contrast, vehicles with lower degree of boxiness appear flatter and sleeker. Notably, *Boxiness* does not align closely with length or wheelbase, suggesting it captures more of a cabin "uprightness" or silhouette shape rather than overall vehicle size. Meanwhile, *Grille Height* exhibits near-zero correlations with physical measures, implying that it reflects primarily stylistic design choices (e.g., tall vs. short front grilles) rather than size. Finally, *Grille Width* is only weakly positively correlated with physical measurements like wheelbase, length, and width. This overall pattern of limited redundancy between visual and functional characteristics is consistent with evidence that consumers trade off visual form and functional attributes (Kang et al. 2019).

Table 6 Correlations: Visual Characteristics vs. Vehicle Dimensions

	Wheelbase	Weight	Length	Height	Width	Height/Width
Body Shape	0.30	0.33	0.39	−0.28	0.25	−0.42
Boxiness	0.05	−0.07	0.14	−0.59	0.02	−0.49
Grille Height	0.04	0.02	0.05	−0.04	0.03	−0.05
Grille Width	0.12	0.08	0.12	0.03	0.15	−0.09

Notes: Entries are pairwise correlations in the estimation sample.

4.3. Incorporating Visual Characteristics in Model of Market Equilibrium

We now map the general demand and supply framework of Section 3.2 to the specific variables in our data. On the demand side, the functional characteristics are $\mathbf{x}_{jt}^D = (\text{hpwt}_{jt}, \text{MP£}_{jt}, \text{space}_{jt})$, and the visual characteristics are $\mathbf{v}_{jt} = (\text{bodyshape}_{jt}, \text{grilleheight}_{jt}, \text{boxiness}_{jt}, \text{grillewidth}_{jt})$. On the cost side, marginal cost depends on $\mathbf{x}_{jt}^C = (\log(\text{hpwt}_{jt}), \log(\text{space}_{jt}), \log(\text{CO}_{2jt}))$ and a time trend contained in $\mathbf{w}_{jt}$. The learned visual characteristics $\mathbf{v}_{jt}$ enter demand (as characteristics and in constructing BLP-style instruments but are excluded from the cost equation and the supply instrument set Z_{jt}^S.

We allow for unobserved heterogeneity in preferences over a selected subset of characteristics via random coefficients σ_{β_1} and σ_{β_2}. As described in Section 3.2, we fix $\sigma_{\text{price}} = 0$ so that all heterogeneity in price sensitivity operates through the income interaction $-\alpha p_{jt}/y_{it}$.

We estimate the model jointly using both demand- and supply-side moment conditions as described in Section 3.2, with the two-step procedure using approximate optimal instruments following Conlon and Gortmaker (2020).

4.4. Estimation Results

Table 7 presents the parameter estimates. Panel A reports the demand-side parameters: the mean preference coefficients ($\overline{\beta}$), the standard deviations of the random coefficients (σ), and the demographic interaction (π) of price with the inverse of income. Panel B reports the supply-side parameters from the log marginal cost specification.

Panel A reports two demand specifications: a baseline model with functional characteristics only and an augmented model that additionally includes four learned visual characteristics. Across both specifications, the functional characteristics have the expected signs and are precisely estimated, and their qualitative interpretation is unchanged when visual characteristics are introduced.

Table 7 BLP Parameter Estimates

	(1) No visual characteristics		(2) With visual characteristics	
	Mean ($\bar{\beta}$)	Std. Dev. (σ)	Mean ($\bar{\beta}$)	Std. Dev. (σ)
Panel A: Demand				
Structured characteristics				
Constant	-13.954***	—	-13.972***	—
	(1.104)		(1.416)	
HP/Weight	2.163***	0.000	2.657***	0.000
	(0.476)	(0.000)	(0.572)	(0.000)
MP£	0.248***	0.000	0.249***	0.009
	(0.044)	(0.368)	(0.063)	(0.425)
Space	0.890***	0.206	0.926**	0.388
	(0.271)	(0.169)	(0.423)	(0.236)
Visual characteristics				
Body Shape	—	—	0.329***	0.000
			(0.094)	(0.000)
Grille Height	—	—	0.216***	0.278*
			(0.069)	(0.165)
Boxiness	—	—	-0.143*	0.472**
			(0.079)	(0.200)
Grille Width	—	—	0.099	0.308
			(0.112)	(0.976)
Price × 1/Income	-136.688***		-179.667***	
	(34.417)		(46.897)	
Panel B: Supply				
Constant	-3.638***		-3.296***	
	(0.228)		(0.169)	
log(HP/Weight)	0.639***		0.644***	
	(0.047)		(0.045)	
log(Space)	1.889***		1.774***	
	(0.089)		(0.073)	
log(CO_2)	0.465***		0.416***	
	(0.082)		(0.074)	
Trend	0.014***		0.013***	
	(0.004)		(0.003)	

Notes: Standard errors in parentheses. † denotes parameters fixed at zero. Stars denote significance using a normal approximation: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

In the augmented model, visual design contributes meaningfully to utility: several learned visual characteristics enter significantly, indicating that consumers value specific front-end design features above and beyond standard performance and size attributes. The model also reveals substantial heterogeneity in design tastes, particularly along the boxiness dimension, suggesting that substitution patterns depend not only on similarity in functional attributes but also on similarity in visual preferences. Finally, the price–income interaction is negative and significant in both specifications, consistent with lower price sensitivity among higher-income households.

Panel B reports the marginal-cost parameters. Estimates are similar across the two specifications, indicating that introducing visual characteristics primarily affects demand rather than the implied cost-side fit.

Figure 3 shows the distributions of implied mean own-price elasticities and markups under both specifications. Including visual characteristics materially increases the magnitude of estimated own-price elasticities. In the distribution across product-years, the mean own-price elasticity shifts from -4.61 (without visual characteristics) to -6.10 (with visual characteristics), with the entire distribution moving left. These magnitudes are comparable to estimates in the automobile demand literature (Berry et al. 1995, Grieco et al. 2021). However, we should expect differences because our data is from the UK market rather than the US market.

We infer marginal costs from the supply-side first-order conditions evaluated at the estimated demand parameters, and compute implied markups as $(p - mc)/p$. Markups are lower under the visual specification, mirroring the increase in price sensitivity. In the distribution, the mean markup declines from 0.24 (without visuals) to 0.18 (with visuals), with the distribution shifting left.

Figure 3 **Estimated Distributions of elasticities and markups with and without visual characteristics.**

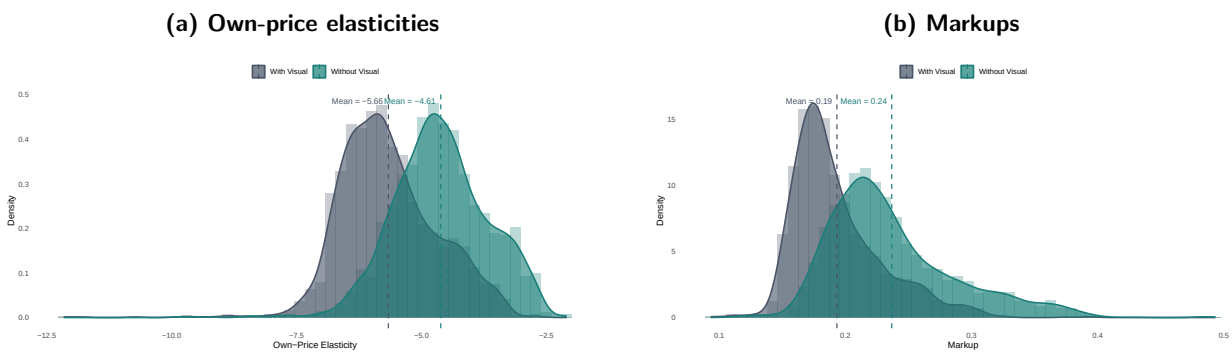
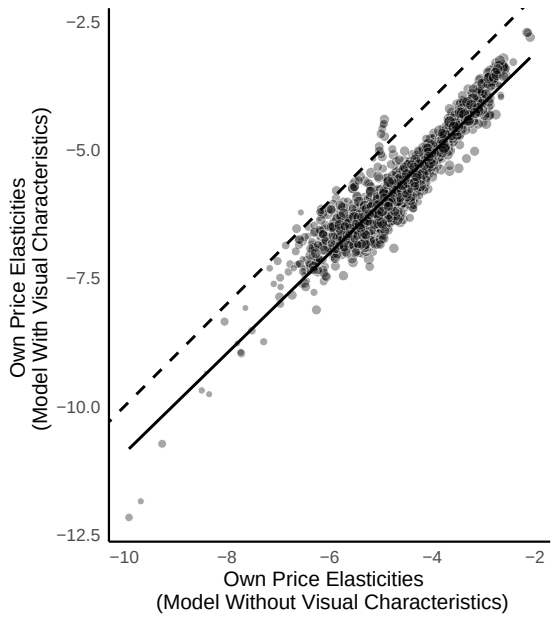


Figure 4 plots the own-price elasticity from the model with visual characteristics against the own-price elasticity from the model without visual characteristics, for each product-year observation. The dashed line represents the 45-degree line: points below it indicate products whose demand became more elastic (more negative) when visual characteristics are included. Nearly all observations (99.4%) fall below this line.

Figure 4 Own Price Elasticities



While the direction of the elasticity shift is nearly universal, the magnitude varies considerably across products. To understand this, we examine whether a product's position in the characteristic space, specifically the density of nearby competitors, predicts the magnitude of the elasticity change.

For each product-year, we define the elasticity ratio as the own-price elasticity from the model with visual characteristics divided by the own-price elasticity from the model without visual characteristics. Since both elasticities are negative, a ratio greater than one indicates that demand became more elastic when visual characteristics were included. We then construct a measure of visual crowdedness: the average Euclidean distance to a product's three nearest neighbors in standardized visual characteristic space (body shape, boxiness, grille height, grille width), computed within each market-year.[5] A smaller average distance indicates a more visually crowded neighborhood, i.e. one with many similar-looking competitors nearby. We construct the analogous measure in standardized functional characteristic space (horsepower-to-weight, miles per pound, space).

4.5. Counterfactual Analysis: Functional vs. Visual Competition

To quantify the competitive role of visual design relative to functional characteristics, we conduct a neighbor-removal counterfactual exercise. For each product j in market $t = 2017$,

[5] We standardize all characteristics to have zero mean and unit variance within each market-year, so that distances are comparable across dimensions and time periods.

we identify its nearest functional neighbor (based on Euclidean distance in the functional characteristic space of hpwt, MP£, and space) and its nearest visual neighbor (based on distance in the disentangled visual characteristic space).[6]

For each focal product, we compute three equilibria: (i) the baseline equilibrium with all products present, (ii) a counterfactual equilibrium with the functional neighbor removed, and (iii) a counterfactual equilibrium with the visual neighbor removed. In each counterfactual, we remove the identified neighbor and re-solve for the Nash–Bertrand pricing equilibrium holding marginal costs fixed, following standard practice in product-removal counterfactuals (e.g., Petrin 2002). We then record the percentage change in the focal product's price, market share, and profit relative to the baseline. This exercise isolates how competitive pressure depends on proximity in each characteristic space. If visual design is an important dimension of product differentiation, then removing a visually similar competitor should relax head-to-head competition and increase the focal product's equilibrium market power, with effects comparable to removing a functionally similar competitor.

Table 8 summarizes mean percentage change across focal products in 2017. Removing the nearest *visual* neighbor generates changes in equilibrium outcomes that are similar in order of magnitude to removing the nearest *functional* neighbor. For prices, the mean adjustment is 0.004% under functional-neighbor removal and 0.009% under visual-neighbor removal. Market share and profit respond in the same direction and are somewhat larger under visual-neighbor removal: the focal product gains 0.556% in market share and 0.576% in profit when its functional neighbor exits, versus 0.639% in market share and 0.696% in profit when its visual neighbor exits.[7]

Table 8 Counterfactual Effects of Product Removal

Outcome	C/F 1: Remove Functional Neighbor	C/F 2: Remove Visual Neighbor
Price (% change)	0.004	0.009
Market Share (% change)	0.556	0.639
Profit (% change)	0.576	0.696

These results highlight that visual similarity can be as competitively consequential as functional similarity. When a visually close competitor is removed, the focal product experiences comparable increases in share and profit as when a functionally close competitor is

[6] Before computing distances, we standardize each characteristic to mean zero and unit variance.

[7] All changes are reported relative to the baseline equilibrium for the focal product.

removed. This suggests that design choices that position a vehicle away from rivals in visual space can soften direct competition in a manner similar to differentiation in performance, operating costs, or size.

Table 9 compares the effects of removing the nearest *functional* neighbor across the two demand specifications. Under functional-neighbor removal, the model with visual characteristics predicts a smaller price response (0.004% vs. 0.007%) but larger gains in market share (0.556% vs. 0.376%) and profit (0.576% vs. 0.409%). These differences indicate that incorporating visual characteristics changes the model-implied reallocation of demand following the exit of a close competitor, which in turn affects the predicted competitive consequences of product positioning.

Table 9 Functional Neighbor Removal: Comparison Across Model Specifications

Outcome	Model Without Visual Characteristics	Model With Visual Characteristics
Price (% change)	0.007	0.004
Market Share (% change)	0.376	0.556
Profit (% change)	0.409	0.576

4.6. Counterfactual Analysis: Economic Value of a Visual Redesign

The neighbor-removal exercise establishes that visual similarity shapes competitive pressure at the market level. We now ask a complementary question at the product level: how much is a specific visual redesign worth?

Automobile manufacturers routinely update the exterior styling of existing models without altering the underlying platform, powertrain, or interior dimensions, a practice known as a *facelift* or mid-cycle refresh. Facelifts typically involve changes to the front fascia, headlamps, grille, and bumper design, while leaving the mechanical and functional attributes of the vehicle unchanged. Because a facelift modifies only the visual design holding functional characteristics fixed, it provides a natural setting for isolating the economic contribution of visual design.

We focus on the Ford C-Max, a compact MPV that received a facelift for the 2016 model year. Figure 5 shows the pre- and post-facelift designs. The redesign substantially altered the front-end appearance: in the learned visual characteristic space, grille height increased by 1.73 standard deviations and grille width increased by 1.57 standard deviations relative to the pre-facelift design, while body shape and boxiness remained largely unchanged.

Figure 5 Ford C-Max: Pre- and Post-Facelift Design

(a) Pre-2016 facelift **(b) Post-2016 facelift**

To quantify the value of this redesign, we conduct a counterfactual exercise: what would have happened to the Ford C-Max's equilibrium price, market share, and profit in 2016 if it had *not* been facelifted? Specifically, we replace the C-Max's 2016 visual characteristics with its 2015 (pre-facelift) values while holding all functional characteristics, marginal costs, and the characteristics of all other products at their observed 2016 levels. We then re-solve for the Nash–Bertrand pricing equilibrium, following the same approach as in the neighbor-removal exercise. The difference between the baseline and counterfactual equilibria isolates the model-implied contribution of the visual redesign to the C-Max's competitive position.

Table 10 reports the results. Under the counterfactual (no facelift), the C-Max's equilibrium price would have been £22 lower (−0.09%), a negligible adjustment reflecting the small price effects typical in differentiated product markets with many competitors. The market share effect is more substantial: without the facelift, the C-Max's market share would have been 6.75% lower. The profit effect is larger still at −33.3%, reflecting the combination of lower share and the high profit leverage inherent in the markup structure.

Table 10 Counterfactual: Ford C-Max Without 2016 Facelift

	Baseline (Actual 2016 Design)	Counterfactual (Keeping 2015 Design)	Δ	$\%\Delta$
Price (in £)	25,263	25,241	−22	−0.09%
Market share ($\times 10^{-4}$)	5.63	5.25	−0.38	−6.75%
Profit (in million £)	57	38	−19	−33.3%

5. Discussion and Conclusion

We explored the integration of visual product characteristics from unstructured image data into structural econometric models, with the goal of determining the economic value of visual design. Our approach is based on disentanglement with supervisory signals, in order to obtain human-interpretable visual characteristics. Alternative approaches in the literature are based on PCA-like methods and do not focus on human interpretability, and rather just attempt to get an uninterpretable low-dimensional representation of the visual design. Without interpretability, we cannot use human judgment to determine which visual characteristics are relevant for demand, and which ones may just be spurious (e.g. color).

There are several potential avenues to extend this research involving visual characteristics in the future. First, we could study the impact of a visually new category like electric cars, which do not achieve significant sales in our dataset. Second, it would be informative to policymakers to understand how the nature of competition in visual space could impact merger outcomes, e.g. firms with different functional characteristics but similar visual characteristics merging might not be viewed as positively as earlier. Third, after modeling engineering and production constraints and incorporating them into a model, it would be helpful to understand how firms can differentiate more in the visual space, and how to coordinate such differentiation with their decisions on functional characteristics. Finally, there is an important and interesting question of investigating consumer choice processes, e.g. consideration, in terms of how individuals integrate information about visual and functional characteristics.

Our results overall point to the important, yet understudied, role of visual design and visual characteristics in determining market outcomes in the automobile market. We would anticipate similar and nuanced effects in several other categories, e.g. furniture, housing etc. where visual design is well known to influence consumer choices.

References

Bengio Y, Courville A, Vincent P (2013) Representation learning: A review and new perspectives. *IEEE transactions on pattern analysis and machine intelligence* 35(8):1798–1828.

Berry S, Levinsohn J, Pakes A (1995) Automobile prices in market equilibrium. *Econometrica: Journal of the Econometric Society* 841–890.

Berry S, Levinsohn J, Pakes A (2004) Differentiated products demand systems from a combination of micro and macro data: The new car market. *Journal of political Economy* 112(1):68–105.

Berry ST (1994) Estimating discrete-choice models of product differentiation. *The RAND Journal of Economics* 242–262.

Bloch PH (1995) Seeking the ideal form: Product design and consumer response. *Journal of marketing* 59(3):16–29.

Buechel EC, Townsend C (2018) Buying beauty for the long run:(mis) predicting liking of product aesthetics. *Journal of Consumer Research* 45(2):275–297.

Burgess C, Higgins I, Pal A, Matthey L, Watters N, Desjardins G, Lerchner A (2017) Understanding disentangling in β-vae. *Workshop on Learning Disentangled Representations at the 31st Conference on Neural Information Processing Systems*.

Chen RTQ, Li X, Grosse RB, Duvenaud DK (2018) Isolating sources of disentanglement in variational autoencoders. *Advances in Neural Information Processing Systems*, 2615–2625.

Compiani G, Morozov I, Seiler S (2025) Demand estimation with text and image data. *arXiv preprint arXiv:2503.20711* .

Conlon C, Gortmaker J (2020) Best practices for differentiated products demand estimation with pyblp. *The RAND Journal of Economics* 51(4):1108–1161.

Creusen ME, Schoormans JP (2005) The different roles of product appearance in consumer choice. *Journal of product innovation management* 22(1):63–81.

Dotson JP, Beltramo MA, Feit EM, Smith RC (2016) Modeling the effect of images on product choices .

Duan S, Matthey L, Saraiva A, Watters N, Burgess C, Lerchner A, Higgins I (2020) Unsupervised model selection for variational disentangled representation learning. *International Conference on Learning Representations*.

Dumoulin V, Visin F (2016) A guide to convolution arithmetic for deep learning. *arXiv preprint arXiv:1603.07285* .

Feenstra RC, Inklaar R, Timmer MP (2015) The next generation of the penn world table. *American economic review* 105(10):3150–3182.

Gandhi A, Houde JF (2019) Measuring substitution patterns in differentiated-products industries. Technical report, National Bureau of Economic Research.

Grieco PL, Murry C, Yurukoglu A (2021) The evolution of market power in the us auto industry. Technical report, National Bureau of Economic Research.

Higgins I, Matthey L, Pal A, Burgess C, Glorot X, Botvinick M, Mohamed S, Lerchner A (2017) beta-vae: Learning basic visual concepts with a constrained variational framework. *International Conference on Learning Representations*.

Homburg C, Schwemmle M, Kuehnl C (2015) New product design: Concept, measurement, and consequences. *Journal of marketing* 79(3):41–56.

Huang J, Chen B, Luo L, Yue S, Ounis I (2021) Dvm-car: A large-scale automotive dataset for visual marketing research and applications. *arXiv preprint arXiv:2109.00881* .

Kang N, Ren Y, Feinberg F, Papalambros P (2019) Form + function: Optimizing aesthetic product design via adaptive, geometrized preference elicitation. *arXiv preprint arXiv:1912.05047* .

Kim H, Mnih A (2018) Disentangling by factorising. *ICML*, 2649–2658.

Kingma DP, Ba J (2014) Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* .

Kingma DP, Welling M (2014) Auto-encoding variational bayes. *International Conference on Learning Representations*.

Krizhevsky A, Sutskever I, Hinton GE (2012) Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems* 25.

Lancaster KJ (1966) A new approach to consumer theory. *Journal of Political Economy* 74(2):132–157.

LeCun Y, Bengio Y, et al. (1995) Convolutional networks for images, speech, and time series. *The handbook of brain theory and neural networks* 3361(10):1995.

Lee K (2025) Generative brand choice .

Liu L, Dzyabura D, Mizik N (2020) Visual listening in: Extracting brand image portrayed on social media. *Marketing Science* 39(4):669–686.

Locatello F, Bauer S, Lučić M, Rätsch G, Gelly S, Schölkopf B, Bachem OF (2019) Challenging common assumptions in the unsupervised learning of disentangled representations. *International Conference on Machine Learning*, 4114–4124.

Locatello F, Tschannen M, Bauer S, Rätsch G, Schölkopf B, Bachem O (2020) Disentangling factors of variations using few labels. *International Conference on Learning Representations*.

Maas AL, Hannun AY, Ng AY, et al. (2013) Rectifier nonlinearities improve neural network acoustic models. *Proc. icml*, volume 30, 3 (Atlanta, GA).

Magnolfi L, McClure J, Sorensen A (2025) Triplet embeddings for demand estimation. *American Economic Journal: Microeconomics* 17(1):282–307.

Nevo A (2000) Mergers with differentiated products: The case of the ready-to-eat cereal industry. *The RAND Journal of Economics* 395–421.

Norman D (2013) *The design of everyday things: Revised and expanded edition* (Basic books).

Petrin A (2002) Quantifying the benefits of new products: The case of the minivan. *Journal of political Economy* 110(4):705–729.

Quan TW, Williams KR (2019) Extracting characteristics from product images and its application to demand estimation. *University of Georgia, Department of Economics* .

Rosen S (1974) Hedonic prices and implicit markets: product differentiation in pure competition. *Journal of political economy* 82(1):34–55.

Sisodia A, Burnap A, Kumar V (2025) Generative interpretable visual design: Using disentanglement for visual conjoint analysis. *Journal of Marketing Research* 62(3):405–428.

Sudhir K (2001) Competitive pricing behavior in the auto market: A structural analysis. *Marketing Science* 20(1):42–60.

Talke K, Salomo S, Wieringa JE, Lutz A (2009) What about design newness? investigating the relevance of a neglected dimension of product innovativeness. *Journal of product innovation management* 26(6):601–615.

Veryzer Jr RW (1993) Aesthetic response and the influence of design principles on product preferences. *Advances in Consumer research* 20(1).

Electronic Companion Supplement

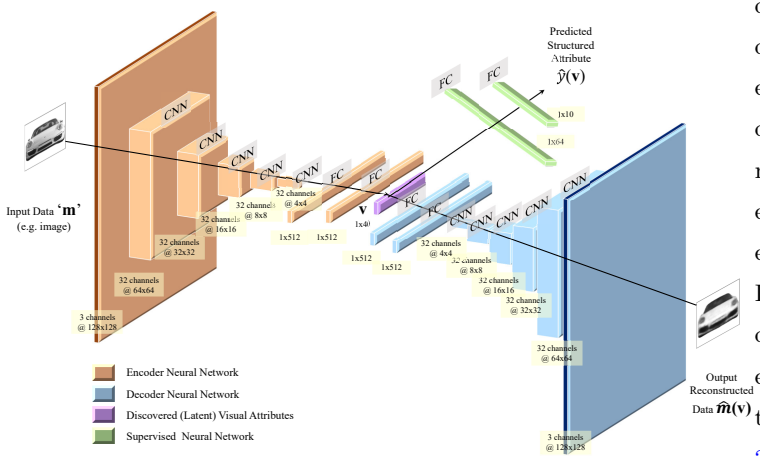
Appendix A: Neural Net Architecture

Figure EC.1 shows the detailed neural net architecture of our model. We modify the architecture proposed by Burgess et al. (2017) to accommodate 128×128 pixel images and incorporate the supervisory signals from our model of market equilibrium.

The encoder neural network uses a sequence of convolutional neural network (CNN) layers to learn high-level representations of the input images. CNNs are well-suited for working with image data, as they can effectively capture spatial hierarchies and learn translation-invariant features (LeCun et al. 1995, Krizhevsky et al. 2012). We stack multiple CNN layers in the encoder to progressively learn more complex and abstract visual concepts. The output of the final CNN layer is then flattened and passed through two fully-connected (FC) layers. The first FC layer reduces the dimensionality of the flattened representation, while the second FC layer further compresses the information into a compact set of latent visual characteristics, with a maximum of J dimensions.

The decoder neural network is designed to reconstruct the original image from the latent visual characteristics. Its architecture is essentially the transpose of the encoder network, consisting of FC layers followed by a sequence of transposed convolutional layers (Dumoulin and Visin 2016). The decoder takes the J-dimensional latent visual characteristics as input and gradually upsamples and expands the representation until it reaches the original image size of 128×128 pixels. Finally, the supervised neural network takes the discovered visual characteristics as input and predicts the vector of supervisory signals, which serve as labels for training the model. The supervised network allows the model to learn visual characteristics that are predictive of the supervisory signals, guiding the disentanglement process.

Figure EC.1 Model Architecture



Notes: The encoder neural net for the VAEs consisted of 5 convolutional layers, each with 32 channels, 4×4 kernels, and a stride of 2. This was followed by 2 fully connected layers, each of 512 units. The latent distribution consisted of one fully connected layer of 40 units parameterizing the mean and log standard deviation of 20 Gaussian random variables. The decoder neural net architecture was the transpose of the encoder neural net but with the output parameterizing Bernoulli distributions over the pixels. Leaky ReLU activations were used throughout, which help alleviate the vanishing gradient problem and improve the model's ability to learn complex representations (Maas et al. 2013). We used the Adam optimizer (Kingma and Ba 2014) with the learning rate 5e-4 and parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$. We set batch size equal to 64. We train for 200 epochs to ensure convergence.

Appendix B: UDR Algorithm

The Unsupervised Disentanglement Ranking (UDR) metric, proposed by Duan et al. (2020), assesses the similarity of disentangled representations learned by different models. The key idea behind UDR is that if two models have learned similar disentangled representations, their informative latent dimensions should exhibit strong correlations.

To compute UDR for a pair of models i and j, we first calculate the correlation matrix R, where each entry $R(a,b)$ represents the correlation between latent dimensions a and b from models i and j, respectively (Equation EC.1). We then identify the most correlated latent dimension in model j for each dimension a in model i, denoted as r_a (Equation EC.2).

$$R(a,b) = cor(v_i(a), v_j(b)) \tag{EC.1}$$

$$r_a = \max_{b \in V(j)} corV(a,b) \tag{EC.2}$$

The UDR score for the pair of models, UDR_{ij}, is computed using Equation EC.3. This equation consists of two symmetric terms, each focusing on one model. The first term considers each informative latent dimension b in model j and calculates the ratio of the squared correlation of its most similar dimension in model i (r_b^2) to the sum of correlations between b and all dimensions in model i. This ratio is then multiplied by

an indicator function $I_{KL}(b)$, which equals 1 if dimension b is informative (determined by its KL divergence from the prior distribution) and 0 otherwise. The second term follows the same process, but with the roles of models i and j reversed.

$$UDR_{ij} = \frac{1}{d_i + d_j} \left[\sum_{b \in Z(j)} \frac{r_b^2}{\sum_{a \in Z(i)} R(a,b)} I_{KL}(b) + \sum_{a \in Z(i)} \frac{r_a^2}{\sum_{b \in Z(j)} R(a,b)} I_{KL}(a) \right] \tag{EC.3}$$

The final UDR score is normalized by the total number of informative dimensions in both models $(d_i + d_j)$ to ensure that having more informative dimensions does not automatically lead to a higher score. A perfect one-to-one correspondence between the informative dimensions of the two models would result in a UDR score of 1.

Appendix C: Validation Surveys for Interpretability and Quantification

To validate the interpretability of the discovered visual characteristics, we conducted a survey with 93 respondents after removing those who failed attention checks. Respondents were shown a sequence of five images for each visual characteristic, where the characteristic varied from left to right while keeping all other characteristics fixed. They were asked to describe how the car changed the most across the sequence of images. Figure EC.2 presents an example of the open-ended question posed to the respondents.

We then used two large language models to summarize the main themes that respondents agreed upon for each visual characteristic.

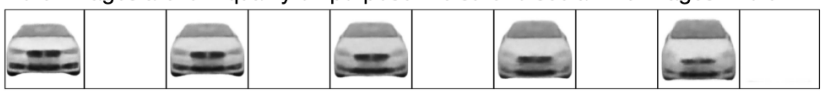
Both language models agreed on the summaries, leading us to label the visual characteristics as follows:

1. Body Shape: The LLM summary of survey respondents states that the "car appears to change in shape, particularly becoming narrower, less angular, and more rounded with each successive image."
2. Color: Automobiles scoring low on this characteristic are lighter and vice-versa.
3. Grille Height: The LLM summary of survey respondents indicates that "grilles are become larger, darker, and more defined." Although this summary also mentions the windscreen becoming darker, which is entangled with the grille height, it captures the essence of the characteristic. We interpret this as automobiles scoring low on this characteristic have less prominent grilles, while those scoring high have more prominent, and larger grilles.
4. Boxiness: The LLM summary of survey respondents describes that the "car becomes lower, flatter, and wider as the sequence progresses." We interpret this as automobiles scoring low on this characteristic have a high degree of boxiness, characterized by a taller, more upright, and narrower shape. In contrast, those scoring high on this characteristic have a lower degree of boxiness, with a lower, flatter, and wider appearance.
5. Grille Width: The LLM summary of survey respondents states that "grille width become bigger, wider, and more pronounced as the sequence progresses." Based on this, we interpret that automobiles scoring high on this characteristic have a wider, more prominent grille, while those scoring low have a narrower, less pronounced grille.

Figure EC.2 Interpretability Validation Survey Question

Q1/4: Look at the below image to see the various parts of a car.

Now, carefully examine each car image below from 1 to 5, going from left to right.
Note: Images are low-quality on purpose. Be sure to see all the images 1 to 5.



1 2 3 4 5

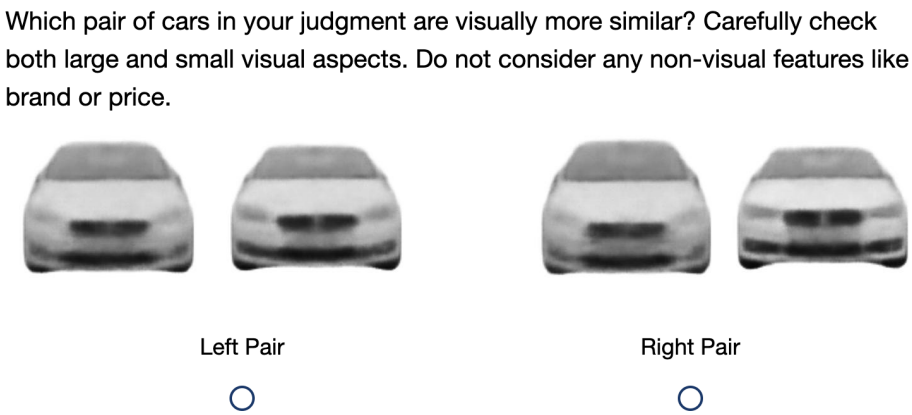
LEFT ⟷ RIGHT

How does the car change the <u>most</u> as you go from image 1 to 5? Go through each part of the car one by one before deciding your response. Write it in a few words.

To further validate the quantification of the visual characteristics determined by our method, we conducted a second survey (Figure EC.3). In this survey, we presented respondents with several pairs of automobile images that differed only along one visual characteristic. Respondents were asked to select the pair of automobiles that they perceived as more similar. We then compared the responses to our algorithm's quantification to assess consistency with human interpretation. For the characteristic we labeled as "Body Shape," 97% of the 104 respondents agreed with the algorithm's quantification scale. Similarly, for "Grille Height," 98% of the 107 respondents were in agreement. The characteristic we termed "Boxiness" had a 95% agreement rate among 103 respondents, while "Grille Width" saw 93% of the 104 respondents concurring with the algorithm's quantification. Overall, a strong majority of respondents (averaging 96% across four visual characteristics) agreed with the algorithm's quantification scale for the visual characteristics, demonstrating that our method's quantification aligns well with human perception.[8]

[8] Agreement is defined as choosing the image-pair that is closer according to the ordering implied by the learned characteristic scores.

Figure EC.3 Quantification Validation Survey Question



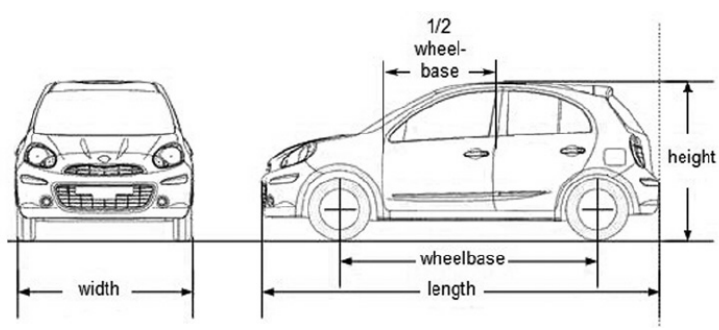
Appendix D: Supervision Signals and UDR-Based Model Selection

This appendix provides additional details on the supervisory signals considered for supervised disentanglement and the corresponding UDR comparisons. UDR lies in $[0, 1]$, with larger values indicating more disentangled representations. Details of the UDR algorithm are provided in Appendix B.

D.1. Candidate Physical Dimensions

Figure EC.4 illustrates the physical dimensions referenced by a subset of supervisory signals used in the disentanglement exercise.

Figure EC.4 Automobile Dimensions Used as Candidate Supervisory Signals



Notes: Figure adapted from **?**.

D.2. Full UDR Comparison Across Supervisory Specifications

Table EC.1 reports the full set of UDR comparisons across alternative supervisory signals (single signals and selected combinations), including unsupervised baselines.

Table EC.1 Comparison of Different Supervisory Approaches (Full Results)

# Signals	Supervisory Signals	β	δ	UDR
3	Wheelbase, Width, Height	50	10	0.739
1	Price	50	30	0.708
1	Weight	50	40	0.708
1	Wheelbase	50	30	0.690
1	Width	50	5	0.689
3	Length, Width, Height	50	40	0.678
1	Length	50	40	0.666
0	Unsupervised β-TCVAE	50	0	0.658
1	Height	30	20	0.378
0	Unsupervised VAE	1	0	0.073
0	Unsupervised AE	0	0	0.074
1	Make	1	1	0.072

Notes: UDR lies in $[0,1]$, with larger values indicating more disentangled representations. For each supervisory specification, we report the (β,δ) pair selected via cross-validated supervised loss over the grid $\beta \in \{1,5,10,20,30,40,50\}$ and $\delta \in \{0,1,5,10,20,30,40,50\}$. Unsupervised baselines correspond to $\delta = 0$. MPG is used only for the supervision comparisons in representation learning and is not used in the structural demand estimation.